Achintya Mathur
23593

Modelling Non-Additive Effects in FUS Phase Separation Using an Extended Flory-Huggins Framework

Biomolecular phase separation of intrinsically disordered proteins such as FUS is strongly influenced by solution conditions, particularly salt concentration and molecular crowding agents like polyethylene glycol (PEG). While the individual effects of salt and PEG on liquid–liquid phase separation (LLPS) have been studied and modelled previously, their cross-interactions remain largely unexplored. This project aims to develop a minimal theoretical framework to test whether the combined influence of salt concentration and crowders like PEG is additive or whether non-trivial coupling emerges.

I will model this system as a 3-component system containing FUS (p), PEG (c), and water (w), as an extended Flory-Huggins free energy functional with concentration-dependent interaction parameters. Salt is treated as an external control parameter that affects interaction via the ionic strength I. The polymer-polymer interactions are modelled as

$$\chi_{pp} = \chi_{pp}^{0} - A\sqrt{I} + BI$$

Where the first term captures the intrinsic protein-protein interactions of FUS. The second term arises from Debye-Hückel screening of electrostatic interactions, and the third term comes from higher-order ion effects. This form is sufficient to produce reentrant phase behaviour, where the system phase separates at low salt, redissolves, and phase separates again with increasing salt concentration.

The influence of PEG will be incorporated through an effective crowding-based interaction of the form

$$\chi_{pc} = K_1\phi_c + K_2\phi_c^2 + \delta I\phi_c$$

Where the first two terms model the concentration-dependent depletion interaction of PEG. The third term is the salt-crowder coupling term, which can be tuned to investigate whether any non-additive effects are observed. Therefore, the total free energy functional is

$$f = \frac{\phi_p}{N_p}\ln\phi_p + \frac{\phi_c}{N_c}\ln\phi_c + \left(1-\phi_p-\phi_c\right)\ln\left(1-\phi_p-\phi_c\right)$$
$$+\left[\chi_{pp}^{0} - A\sqrt{I} + BI\right]\phi_p^2 + \left[K_1\phi_c + K_2\phi_c^2 + \delta I\phi_c\right]\phi_p$$

Phase behaviour will be determined by the binodal conditions. The binodal is defined by equating the chemical potentials and osmotic pressures in the dense and dilute phases, equivalent to a common tangent construction.

$$\left.\frac{\partial f}{\partial \phi_p}\right|_{\phi_1} = \left.\frac{\partial f}{\partial \phi_p}\right|_{\phi_2} = \frac{f(\phi_2) - f(\phi_1)}{\phi_2 - \phi_1}$$

These conditions will be solved numerically to generate phase diagrams as a function of protein, PEG, and salt concentration. Model predictions will be compared with experimental data from PhaseScan, a high-throughput microfluidics platform that provides LLPS boundaries for a wide range of conditions.

A stepwise fitting procedure will be used. Salt-only data will be used to fit A and B, while PEG-only data will be used to fit K_1 and K_2. Combined data will be used to estimate δ. A non-zero value would provide evidence for non-additive behaviour.

This model is meant to be minimal, which means it has many limitations. It treats FUS as a homogeneous polymer, neglecting any sequence-specific interactions or local charges in the protein. The implicit treatment of salt, used to simplify the model, assumes negligible salt partitioning between the two phases and ignores ion-specific effects.

References

1. Pappu, R. V., Cohen, S. R., Dar, F., Farag, M., & Kar, M. (2023). Phase transitions of associative biomacromolecules. *Chemical reviews*, *123*(14), 8945-8987.
2. Arter, W. E., Qi, R., Erkamp, N. A., Krainer, G., Didi, K., Welsh, T. J., ... & Knowles, T. P. (2022). Biomolecular condensate phase diagrams with a combinatorial microdroplet platform. *Nature Communications*, *13*(1), 7845.
3. Mondal, S., & Shakhnovich, E. (2025). The Origin of the Ionic-strength Dependent Reentrant Behavior in Liquid-Liquid Phase Separation of Uncharged IDPs. *bioRxiv*.
4. Yong, H., & Merlitz, H. (2025). An Analytical Understanding of Reentrant Condensation of a Polyelectrolyte in the Presence of an Oppositely Charged Surfactant. *Langmuir : the ACS journal of surfaces and colloids*, *41*(30), 19683–19697.
5. Robles-Hernández, B., González-Burgos, M., Malo de Molina, P., Asenjo-Sanz, I., Radulescu, A., Pomposo, J. A., ... & Colmenero, J. (2023). Structure of single-chain nanoparticles under crowding conditions: A random phase approximation approach. *Macromolecules*, *56*(21), 8971-8979.
6. André, A. A. M., Yewdall, N. A., & Spruijt, E. (2023). Crowding-induced phase separation and gelling by co-condensation of PEG in NPM1-rRNA condensates. *Biophysical journal*, *122*(2), 397–407. https://doi.org/10.1016/j.bpj.2022.12.001